VF Asia Footwear Laboratory Testing Manual

Test Method for Corrosion Resistance	Test method # VF- E09 REF # : TM-R- 010 Page 1 of 2
Date Issued : December 20, 2014 Revision # : 1 Reasons for revision: 1. Change the concentration of sodium chloride water solution from 1 % to 3 % 2. Change reference number NOTE: *Timberland adopt VF test method - specimen partially immersed in 3% NaCl with crockmeter cloth partially cover with immersion dish.*	Created by : Nila Babia Erazo / Chris Adam

PURPOSE AND SCOPE: To measure the ability of metal components to resist corrosion due to water exposure.

REFERENCE: ASTM G44

EQUIPMENT AND MATERIAL :

1. Immersion dish, which is large enough to contain the test sample such that no portion contacts the sides of the dish.
2. Sodium Chloride (NaCl) – AR Grade
3. Crock meter test cloth ,large enough to place all test samples onto it such that no two test samples contact each other
4. Glass plate, at least as large as the opening of the immersion.
5. Metal components (eyelets, hooks, buckles, spikes, etc.).

SAMPLE PREPARATION:

1. Condition the sample for at least 24 hours at the standard laboratory condition at 23.0±2.0 ˚C temperature and 55 ±10 % Relative Humidity.
2. Use test materials which have been completely separated from any other portion of the shoe (i.e., utilize the metal component only).

REPETITIONS: Test two (2) samples per material**.**

TEST PROCEDURE:

1. Place a piece of the white cloth onto the bottom of the immersion dish.
2. Place the test samples on the white cloth, such that no two test samples contact each other or the sides of the immersion dish.
3. Prepare a 3% sodium chloride water solution. To prepare 3%, weigh 3 grams of NaCl , dissolve in 97 ml of distilled water.
4. Pour the 3% NaCl solution into the immersion dish until the test samples is approximately half immersed. Do not completely immerse the test samples, Record the time.

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5. Place the glass plate on top of the immersion dish such that a small opening remains (the opening should be approximately 1/10 the surface area of the opening of the immersion dish).

6. Examine the samples at least every two (2) hours during the working day, record any changes in appearance to the test sample, including signs of rust/corrosion, peeling of plating, discoloration, tarnishing etc.

7. After 24 hours, replace any water which has evaporated from the salt solution (so that the test materials remain approximately half immersed).
8. Conclude the test after 48 hours

Note: The failure point is defined as the time when any signs of rust / corrosion occur or any removal of the plating occurs. The sample can also fail if discoloration and or tarnishing is to an unacceptable level.

REPORT: Any change in the appearance of the test sample or discoloration of the white cloth, including the time (to the nearest 0.5 hours) at which the change /discoloration began to occur.